
SECOND-ORDER SUPPORT CERTIFICATION FOR IN-EXACT REGULARIZED OPTIMIZATION

Anonymous authors

Paper under double-blind review

ABSTRACT

Support decisions in sparsity-regularized learning depend on optimality inequalities that can remain unresolved even when an optimization residual is small. We study certification from the value, gradient, and Hessian at one inexact candidate, together with a valid neighborhood bound on Hessian variation. For a convex twice continuously differentiable loss with positive curvature on the candidate’s active coordinates, we give explicit tests that certify the unique optimizer’s signed support. The certificate first localizes a restricted stationary point, then cancels the first-order error in each inactive gradient through a Newton correction. Its remaining uncertainty is proportional to the square of the active residual. A two-dimensional construction shows a corresponding information limitation: two globally strongly convex smooth losses have identical second-order information and common smoothness bounds at the candidate, yet their optimizers disagree on an inactive coordinate. Every deterministic estimate of the restricted inactive gradient from those data incurs order-two worst-case error. Numerical sweeps illustrate the support ambiguity and the abstention caused by conservative derivative bounds. The results distinguish a sufficient local certificate from the information required to resolve support near an optimality boundary.

1 INTRODUCTION

A sparse iterate is not itself a certificate of a sparse optimum. In an ℓ_1 -regularized problem, a coordinate equal to zero at an intermediate iterate may enter later, while a small nonzero coordinate may eventually disappear. This distinction matters whenever the support is used to reduce a subsequent optimization problem, select a model, or decide whether additional optimization can change the reported variables. The relevant target is the optimizer of the specified regularized objective. Recovering a data-generating support is a separate statistical question.

Guaranteed feature elimination and working-set methods already exploit the structure of sparse optimization (Ghaoui et al., 2010; Ndiaye et al., 2017; Johnson & Guestrin, 2015; Massias et al., 2018). Their success motivates a more localized question. Suppose an algorithm supplies one candidate, its gradient and Hessian, and a justified bound on how the Hessian varies nearby. What signed support can those quantities certify, and what ambiguity remains even when the candidate’s active residual is small? The answer depends on both curvature and the distance of inactive optimality conditions from their thresholds.

Newton prediction is particularly relevant because it approximates the displacement of the active variables. For losses of degree two, the active gradient is affine, so an active Newton solve can recover the restricted optimizer exactly. Hessian screening already uses this affine structure in regularization-path prediction (Larsson & Wallin, 2022). For nonlinear losses, however, the inactive gradient changes with the active displacement. A first-order prediction becomes a certificate only after this change is bounded. Controlling the active coordinates alone leaves the inactive optimality inequalities unresolved.

We derive a sufficient certificate that connects these two tasks. A positive active Hessian and a small dimensionless residual yield a restricted root in an explicit ball. Substituting the re-

stricted root equation into the inactive-gradient expansion removes its leading displacement error. The resulting coordinatewise enclosure has radius proportional to the square of the active residual, with an explicit factor measuring active-to-inactive Hessian coupling. Strict corrected sign and inactive-gradient tests then establish a unique global optimizer. Global strong convexity is unnecessary: strict inactive inequalities force every optimizer onto the same active subspace, where local positive curvature is sufficient for uniqueness.

The distinction from trajectory identification is substantive. Under nondegeneracy, proximal methods can identify an optimal manifold after finitely many iterations (Liang et al., 2015; Nutini et al., 2017). Such results describe a sequence approaching an optimum. The certificate developed below accepts an arbitrary supplied candidate and uses inequalities that can be evaluated without knowing that optimum. Nevertheless, its proof shares the classical localization, Taylor expansion, and optimality machinery of Newton and identification analyses (Lee et al., 2014). The contribution is the explicit support enclosure and its information boundary, rather than a different Newton convergence principle.

A companion construction explains why the second-order remainder cannot simply be deleted. We exhibit two smooth losses in two dimensions with identical values, gradients, and Hessians at the candidate. Both have uniform global strong convexity and the same supplied local Hessian bound. Their restricted optimizers coincide, but their inactive derivatives lie on opposite sides of the ℓ_1 threshold. A deterministic rule observing only the common information must return the same answer for both objectives. Thus an always-correct binary decision is impossible on this pair, and every real-valued inactive-gradient estimate has a order-two worst-case error. This is a single-candidate information statement; additional function or derivative evaluations change the information available.

The numerical study examines this distinction under controlled conditions. Affine-calibrated logistic losses provide known sparse optima and analytic derivative bounds, allowing every successful certificate to be checked. A least-squares lasso subclass compares the zero-remainder Newton test with a basic dual-gap screening sphere, while the polynomial pair displays the support ambiguity directly. These experiments report both certification and abstention. Their purpose is to expose how the sufficient conditions behave, including regimes where correction offers no measured advantage, without treating numerical support as a proof or as a general solver-speed comparison.

2 RELATED WORK

Screening and working sets. Safe screening certifies zero coefficients by enclosing an optimal dual point and bounding the correlations over that region. Foundational feature elimination (Ghaoui et al., 2010), dual-polytope projection (Wang et al., 2012), and geometric screening tests (Xiang et al., 2014) develop this perspective. The Gap Safe framework converts a primal–dual gap into a safe region for sparsity-enforcing penalties (Fercq et al., 2015; Ndiaye et al., 2017). In particular, Proposition 5 and Theorem 6 of Ndiaye et al. (2017) connect a dual feasible point, smooth losses, and a dual-gap radius to guaranteed elimination. Our certificate instead encloses a restricted primal root and then certifies all signs and inactive coordinates together. Both approaches rely on optimality inequalities; their available information and sufficient conditions differ.

Screening can be integrated dynamically into first-order methods (Bonnetfoy et al., 2014), extended to features and samples jointly (Shibagaki et al., 2016), or adapted to approximate structured dictionaries (Dantas & Gribonval, 2018). Local regularity also appears in expanded Gap Safe rules (Dantas et al., 2021), and dynamic Sasvi derives regions through convex duality for norm-regularized least squares (Yamada & Yamada, 2021). These developments preclude interpreting the use of local curvature alone as a new screening principle. The specific object studied below is an inactive-gradient enclosure after a restricted Newton correction, with a fully stated neighborhood requirement on the full Hessian.

Working sets prioritize a subset of variables that are likely to matter; their role differs from certifying that a variable is absent at every optimum. Blitz (Johnson & Guestrin, 2015) and Celer (Massias et al., 2018) combine reduced problems with optimality control,

and Celer improves dual points by extrapolation. The Hessian Screening Rule (Larsson & Wallin, 2022) predicts changes along a regularization path. Its Theorem 3.1 and Remark 3.2 describe the exact affine path for least-squares losses under the relevant active-set and optimality conditions, while its safeguarded algorithm controls the dual gap. The present analysis fixes the regularization parameter and studies nonlinear residual uncertainty at one candidate. Hessian prediction, exactness for restrictions with constant Hessian, and the need for safeguards are shared foundations.

Identification and inexact optimization. Proximal algorithms provide a general language for nonsmooth composite objectives (Parikh & Boyd, 2014). Inexact proximal-gradient analysis controls convergence when gradients or proximal subproblems are computed approximately (Schmidt et al., 2011). Proximal Newton analysis exploits curvature for local rapid convergence (Lee et al., 2014); Theorem 3.4 of that work establishes local convergence with an error proportional to the square of the preceding error under strong convexity and Lipschitz smoothness assumptions. Interior-point methods for sparse logistic regression (Koh et al., 2007) and self-concordant logistic analysis (Bach, 2009) provide additional second-order perspectives. The certificate in this paper is a test applied to an output of such an optimization procedure, rather than a convergence theorem for that procedure.

Finite activity identification uses nondegeneracy to turn convergence into eventual membership of an optimal manifold. Theorem 3.4 of Liang et al. (2015) treats convergent forward-backward-type sequences with a partly smooth regularizer and a nondegenerate optimum. Lemma 1 of Nutini et al. (2017) gives a direct separable-penalty argument using strong convexity, gradient smoothness, and a margin at the optimum. Their quantitative analysis makes the dependence on that margin explicit. Regularized dual averaging (Xiao, 2010) and its manifold-identification analysis (Lee & Wright, 2012) address stochastic candidate sequences. Our deterministic candidate test neither imports these stochastic guarantees nor replaces them: it supplies an independently checkable sufficient condition once a candidate and a valid local derivative bound are available.

Corrected outputs and information limitations. The cancellation used below is related to residual correction for nonlinear outputs. For example, Theorem 8.5 of Ishizaka (2026) combines verified root localization, linearized stability, and output regularity in an adjoint-corrected enclosure for a semilinear PDE problem. Its application and function spaces differ, but the correction-plus-remainder mechanism overlaps with our inactive-gradient calculation. The sparsity-specific step is to convert the enclosure and corrected coordinate signs into full nonsmooth optimality and uniqueness.

Smooth interpolation and oracle indistinguishability are established tools for optimization lower bounds. Theorem 4 of Taylor et al. (2015) characterizes finite value-gradient interpolation for smooth strongly convex functions, while Theorem 1 of Garg et al. (2021) gives a higher-order oracle lower bound for approximate minimization in a high-dimensional setting. Our construction prescribes a common Hessian as well as a common value and gradient, and its output is a support decision in two dimensions. It supplies a concrete local ambiguity with a second-order residual scale. It is neither a stronger multi-query complexity lower bound nor a claim that indistinguishability itself is a new proof method.

3 LOCAL INFORMATION AND SUPPORT MARGINS

Let p be a positive finite integer and let

$$F(z) = f(z) + \lambda \|z\|_1, \quad z \in \mathbb{R}^p, \quad \lambda > 0. \quad (1)$$

All vector norms without a subscript are Euclidean, and matrix norms are induced operator norms. Fix a candidate x with nonempty support $S = \{i : x_i \neq 0\}$ and signs $s = \text{sign}(x_S)$. The coordinate embedding $P : \mathbb{R}^S \rightarrow \mathbb{R}^p$ inserts zeros outside S , so $P^\top P = I$. Subscripts on a Hessian denote its indicated rows and columns. The certificate treats x as given; it imposes no condition on the algorithm that produced it.

Assumption 1 (Convex loss). *The function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is convex and twice continuously differentiable.*

Convexity will turn a verified subgradient equation into global optimality, while differentiability permits an integral remainder along the active subspace.

Write

$$H = \nabla_{SS}^2 f(x), \quad m = \lambda_{\min}(H), \quad r = \nabla_S f(x) + \lambda s, \quad a = \frac{\|r\|}{m}, \quad d = H^{-1}r. \quad (2)$$

The quantities involving m and H^{-1} are used only when H is positive definite. The residual r measures stationarity of the smooth restriction with the candidate signs held fixed. It need not be the smallest-norm full subgradient of F . This choice is deliberate: the inactive inequalities will be checked separately after the restricted root is localized.

Assumption 2 (Neighborhood curvature control). *The active Hessian H is positive definite. A supplied constant $L \geq 0$ satisfies*

$$\|\nabla^2 f(x + Ph) - \nabla^2 f(x + Pk)\|_{\text{op}} \leq L\|h - k\| \quad \text{for all } \|h\|, \|k\| \leq 2a. \quad (3)$$

Moreover, $La/m \leq 1/4$.

The bound concerns the *full* Hessian along active displacements. A bound on $\nabla_{SS}^2 f$ alone would localize the restricted root but would not control inactive gradients. Although the ball depends on the computed residual, its radius is known before the test is applied. The constant must be valid on that entire ball; observing Hessians at finitely many points is not a substitute for this requirement.

For $j \notin S$, define the corrected inactive gradient and its radius by

$$q_j = \partial_j f(x) - \nabla_{jS}^2 f(x)d, \quad (4)$$

$$E_j = 2La^2 \sqrt{1 + \|\nabla_{jS}^2 f(x)H^{-1}\|^2}, \quad b = \frac{2La^2}{m}. \quad (5)$$

The scalar b bounds the error of the active Newton displacement. The factor in E_j records how active residual error propagates into coordinate j . Even a small Taylor remainder may be amplified when the active Hessian is poorly conditioned.

Assumption 3 (Computed strict margins). *The candidate data satisfy*

$$s_i(x_i - d_i) > b \quad (i \in S), \quad |q_j| + E_j < \lambda \quad (j \notin S). \quad (6)$$

These inequalities contain no unknown optimizer. The active test protects the sign after correction; the inactive test puts an entire derivative interval inside the subgradient threshold. If an inequality fails, the procedure abstains. Failure is not evidence that the candidate support is incorrect. Empty inactive sets require no inactive tests, and $L = 0$ is allowed. The case of an empty active set is outside this formulation because the active inverse and eigenvalue would not be defined.

4 A SECOND-ORDER SUPPORT CERTIFICATE

The obstacle is to establish full optimality from a restricted equation without assuming that the proposed support is already correct. A root of the restricted gradient could have different signs, and its inactive gradients could violate the ℓ_1 threshold. The explicit radii in (5) control both possibilities.

Theorem 1 (Signed-support certificate). *Under Assumptions 1–3, the objective (1) has a unique global minimizer z^* . Its support is exactly S and $\text{sign}(z_S^*) = s$. With $h^* = z_S^* - x_S$,*

$$\|h^*\| \leq 2a, \quad \|h^* + d\| \leq b, \quad |\partial_j f(z^*) - q_j| \leq E_j \quad (j \notin S).$$

If $r = 0$, then $z^ = x$ and $h^* = b = E_j = 0$. The complete proof is in Appendix A.1.*

Restricted localization. Let $G(h) = \nabla_S f(x + Ph) + \lambda s$. On the ball $\|h\| \leq 2a$, integrating the Hessian along the segment from x gives $G(h) = r + Hh + R_S(h)$ with

$\|R_S(h)\| \leq L\|h\|^2/2$. The fixed-Hessian Newton map $\Phi(h) = -d - H^{-1}R_S(h)$ therefore satisfies

$$\|\Phi(h)\| \leq a + \frac{2La^2}{m} \leq \frac{3}{2}a, \quad \|D\Phi(h)\|_{\text{op}} \leq \frac{2La}{m} \leq \frac{1}{2}. \quad (7)$$

The first inequality preserves the ball and the second contracts distances. Their combination constructs a restricted root and yields $h^* + d = -H^{-1}R_S(h^*)$, which is the source of the radius b . This is classical Newton localization expressed in candidate quantities; it is not a convergence-rate improvement over proximal Newton analysis (Lee et al., 2014).

Inactive output cancellation. The full remainder is $R(h) = \nabla f(x + Ph) - \nabla f(x) - \nabla^2 f(x)Ph$. At the restricted root, the active equation removes the first-order displacement error:

$$\begin{aligned} \partial_j f(x + Ph^*) - q_j &= R_j(h^*) - \nabla_{jS}^2 f(x)H^{-1}R_S(h^*), \\ |\partial_j f(x + Ph^*) - q_j| &\leq \sqrt{1 + \|\nabla_{jS}^2 f(x)H^{-1}\|^2} \frac{L\|h^*\|^2}{2} \leq E_j. \end{aligned} \quad (8)$$

The square root comes from the norm of one row acting on the *same* full remainder vector. Because $j \notin S$, its inactive unit coordinate is orthogonal to its active block. Bounding the two remainder terms independently would discard this structure. The argument is related to corrected-output enclosures (Ishizaka, 2026); the output is now an inactive optimality derivative.

From the root to the optimizer. The proof has three logical stages. First, (7) establishes a restricted root without any sign premise. Second, the active error radius and (8) verify its signs and strict inactive subgradients. Third, convexity proves global optimality, and strict inactive subgradients force all other minimizers into the active subspace. The local active Hessian remains at least $mI/2$ on the root neighborhood, so a distinct minimizer would create a nonconstant segment on which a locally strictly convex objective was constant. This contradiction proves uniqueness without global strong convexity.

What the radius measures. For fixed m , L , and coupling rows, E_j is proportional to the square of $\|r\|$. The leading inactive-gradient change has already been included in q_j ; the radius measures what the second-order data cannot determine through that linear correction. The theorem requires positive margins after subtracting these uncertainties. It makes no assertion at equality, where a coordinate can be at the threshold even at an optimum.

For a loss of degree two, $L = 0$ and both radii vanish. The restricted correction is exact, reducing the theorem to direct sign and KKT checks. This specialization agrees with the affine gradient structure underlying Hessian screening (Larsson & Wallin, 2022). For nonlinear losses, a smaller residual helps only in conjunction with a valid neighborhood bound. Increasing a valid L preserves soundness but can cause abstention, as the experiments demonstrate. Thus the theorem separates the size of an optimization residual from the strength of the conclusion that can be certified from it.

5 A COMMON-JET INFORMATION BOUNDARY

A second-order upper radius could reflect a conservative proof rather than an intrinsic information loss. To distinguish these possibilities, we keep the supplied second-order data fixed while changing the true inactive derivative. The perturbation must preserve global convexity; an unconstrained signed cubic alone would not do so.

Set $p = 2$, $\lambda = 1$, $x = (1, 0)$, $t = u - 1$, $\kappa = 1/8$, and $B = 9/64$. For $0 < \epsilon \leq 1/10$ and $\sigma \in \{-1, +1\}$, define

$$f_\sigma(u, v) = \frac{t^2 + v^2}{2} + (\epsilon - 1)t + \left(1 - \frac{\epsilon^2}{16}\right)v + \sigma\kappa t^2 v + B(t^2 + v^2)^2. \quad (9)$$

Define $F_\sigma(u, v) = f_\sigma(u, v) + |u| + |v|$. The decision rule receives x, λ , the value, gradient, and Hessian at x , and the common bounds $m = 1$ and $L = 4$ on the radius-1/2 ball around x .

It may also know a global strong-convexity lower bound of $3/4$. The sign σ is not supplied, and evaluations away from x are outside this information model.

Theorem 2 (Common-jet obstruction). *The losses (9) are globally $3/4$ -strongly convex and C^∞ . They have identical second-order jets at x and the common local Hessian Lipschitz bound $L = 4$. The minimizers of F_- and F_+ restricted to $v = 0$ coincide at $(1 + t_\epsilon, 0)$, where*

$$t_\epsilon + 4Bt_\epsilon^3 = -\epsilon, \quad -\epsilon < t_\epsilon < 0, \quad |t_\epsilon| \geq \frac{\epsilon}{1 + 4B\epsilon^2}.$$

At this restricted minimizer, the inactive gradients differ by $2\kappa t_\epsilon^2 \geq \epsilon^2/8$. The unique minimizer of $f_- + \|\cdot\|_1$ has support $\{1\}$, whereas the unique minimizer of $f_+ + \|\cdot\|_1$ has a nonzero second coordinate. Consequently no deterministic rule using only the supplied common information correctly decides inactivity for both objectives. Every real-valued estimate of the restricted inactive gradient from that information has worst-case absolute error at least $\epsilon^2/16$. The complete proof is in Appendix A.2.

Convexity without changing the jet. Write $w = (t, v)$ and $\rho = \|w\|$. The signed cubic has a Hessian with norm at most $6\kappa\rho$, while the quartic contributes a positive semidefinite term with least eigenvalue $4B\rho^2$. Hence

$$\nabla^2 f_\sigma(u, v) \succeq (1 - 6\kappa\rho + 4B\rho^2)I \succeq \left(1 - \frac{9\kappa^2}{4B}\right)I = \frac{3}{4}I. \quad (10)$$

Both perturbations vanish through second order at $w = 0$. They therefore preserve the same supplied jet while allowing opposite signs of the mixed cubic term. The full Hessian derivative is bounded by $6\kappa + 12B = 39/16 < 4$ on the half-radius ball, giving the common L .

Threshold separation. The restricted root equation is scalar and monotone. It implies $t_\epsilon^2 > \epsilon^2/2$, so the inactive derivatives satisfy

$$\begin{aligned} g_\sigma &:= \partial_v f_\sigma(1 + t_\epsilon, 0) = 1 - \epsilon^2/16 + \sigma\kappa t_\epsilon^2, \\ g_+ &> 1, \quad -1 < g_- < 1, \quad g_+ - g_- = 2\kappa t_\epsilon^2 > \epsilon^2/8. \end{aligned} \quad (11)$$

The first two inequalities give the different support decisions. Any estimate q based on the common data is the same for both objectives, and $\max_\sigma |q - g_\sigma| \geq |g_+ - g_-|/2$ gives the stated lower bound.

The proof first verifies uniform convexity and smoothness, then identifies the common restricted optimizer, and finally converts the inactive derivative difference into an information obstruction. Each part matters: matching jets alone says nothing about support, and a support change in a nonconvex construction would not establish the stated boundary for convex learning objectives.

The result is an order comparison in $\epsilon = \|r\|$ with fixed supplied curvature and smoothness, not an equality of optimal constants. It concerns a restricted inactive output even when that restricted point is not the full optimum. In particular, a sound rule may abstain on the common data. The positive theorem is compatible with this limitation because its strict certificate tests need not hold on the pair. For the smaller residuals satisfying $4\epsilon \leq 1/4$, even the localization condition holds, while the inactive uncertainty still prevents the common information from distinguishing the two support outcomes. General higher-order oracle bounds (Garg et al., 2021) allow repeated queries and address a different output; the construction isolates what is lost when those additional queries are unavailable.

6 NUMERICAL CALIBRATION AND SUPPORT DECISIONS

The numerical program tests how the sufficient certificate behaves when the optimizer is known exactly. It contains three controlled families: a nonlinear logistic loss with adjustable inactive margin and curvature, a least-squares lasso comparison, and the common-jet pair. All data are generated offline. Random designs use 20 fixed seeds, and the deterministic polynomial sweep uses 41 residual values. The measurements concern optimization support, not prediction accuracy or statistical variable selection.

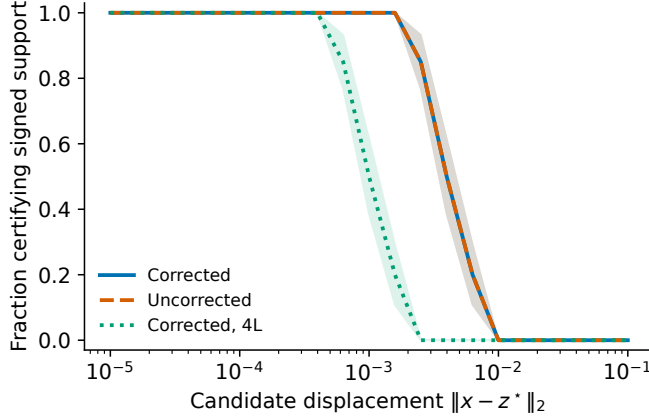


Figure 1: Certificate conservatism on affine-calibrated logistic losses: $n=80$, $p=12$, three nonzero optimal coefficients, Gaussian design correlation .7, ridge .1 and inactive margin .01. Means and one-standard-error bands use 20 seeds. Corrected and uncorrected decisions coincide in this slice; multiplying the analytic Hessian Lipschitz bound by four causes earlier abstention. Candidates are perturbed only on the known active support.

Nonlinear losses with known optima. For $n = 80$, $p = 12$, and $s = 3$, construct design rows $A_{ij} = \sqrt{0.7} Z_{i0} + \sqrt{0.3} Z_{ij}$ from independent standard normal variables. Let $z^* = (1, -0.8, 0.6, 0, \dots, 0)$ and $y_i = \text{sigmoid}(A_i z^*)$. The smooth objective is

$$f(z) = \frac{1}{n} \sum_{i=1}^n \{\log(1 + \exp(A_i z)) - y_i A_i z\} + \frac{\mu}{2} \|z\|^2 + c^\top z.$$

With $\lambda = 1$, choose $c_S = -\text{sign}(z_S^*) - \mu z_S^*$ and $c_j = 1 - \delta$ off S . The logistic part has zero gradient at z^* , so these choices give exact active stationarity and inactive slack δ . Positive μ makes the optimum unique. The responses are soft labels and the affine term calibrates the optimum; this construction is a controlled convex test rather than an observational classification dataset.

The scalar logistic third derivative has absolute value at most $1/(6\sqrt{3})$. Therefore the full Hessian has the global Lipschitz bound $L = n^{-1} \sum_i \|A_i\|^3 / (6\sqrt{3})$, since each third-derivative contribution is bounded by its scalar coefficient times $\|A_i\|^3$. We use this analytic bound, not a fitted estimate. For each seed, perturb the active coordinates along one normalized Gaussian direction, at 21 logarithmically spaced displacements from 10^{-5} to 10^{-1} . Sweep $\mu \in \{0.001, 0.1\}$ and $\delta \in \{0.001, 0.01, 0.1\}$, giving 2,520 candidates.

The uncorrected comparison uses the same restricted-root and sign conditions, but encloses the inactive derivative around $\partial_j f(x)$ with radius $U_j = 2a \|\nabla_{jS}^2 f(x)\| + 2La^2$. This follows by bounding the linear displacement and Taylor remainder separately on the same ball. A second control multiplies the valid L by four in every part of the corrected test. Both controls remain sufficient tests; they measure different sources of conservatism.

Across the full grid, the corrected, uncorrected, and inflated- L tests certify 1,383, 1,322, and 1,023 candidates, respectively. No successful corrected certificate violates the known optimizer’s output or displacement enclosure beyond 10^{-9} . These counts describe the declared grid; they are not estimates of performance on arbitrary learning problems. Figure 1 shows the $\mu = 0.1$, $\delta = 0.01$ slice. The corrected and uncorrected decisions coincide in that slice, while conservative smoothness causes earlier abstention. Thus removing a leading output-error term need not change the final binary decision when another sufficient condition is limiting.

Quadratic control and dual-gap screening. For each of 20 additional seeds, draw a 24×12 Gaussian design with entries scaled by $1/\sqrt{24}$ and use the same three-coordinate optimizer. Choose a residual e solving $A^\top e = -g^*$, where $g_S^* = -\text{sign}(z_S^*)$ and $g_j^* = 0.99$ off

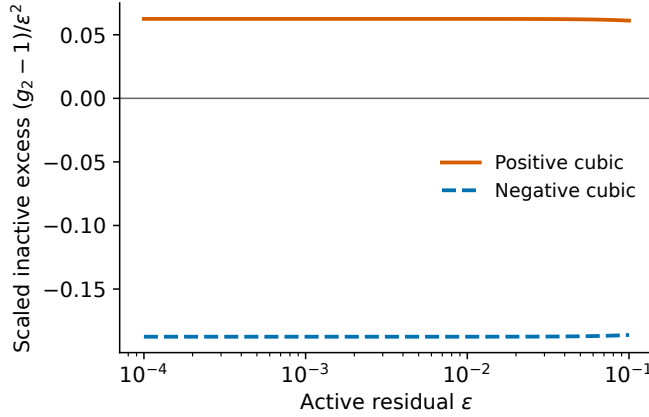


Figure 2: Opposite inactive decisions from a common second-order jet. For the two-dimensional polynomial pair with $\kappa=1/8$ and $B=9/64$, the curves show the inactive derivative minus the threshold 1, divided by the active residual squared, at 41 logarithmically spaced residuals. The solid positive-cubic curve lies above zero and the dashed negative-cubic curve below it. The sweep is deterministic; no sampling uncertainty applies.

S , and set $y = Az^* + e$. The resulting lasso objective is $\|Az - y\|^2/2 + \|z\|_1$. Perturbations on the known support again use the 21-point displacement grid.

Because the Hessian is constant, the restricted Newton test has zero remainder and certifies all 420 candidates. For comparison, form the feasible dual point $\theta = (y - Ax)/\max\{1, \|A^\top(y - Ax)\|_\infty\}$. The dual objective is $D(\theta) = y^\top\theta - \|\theta\|^2/2$, the gap is $G = F(x) - D(\theta)$, and the safe sphere test is $|A_j^\top\theta| + \|A_j\|\sqrt{2G} < 1$ (Fercoq et al., 2015; Ndiaye et al., 2017). This sphere eliminates a mean fraction 0.18968 of the nine inactive coordinates across the fixed grid. The comparison benefits the Newton test by supplying the correct candidate support and a restriction of degree two. It uses neither improved dual extrapolation nor a production working-set implementation. The two numbers compare certificate outputs on constructed inputs, not total solver runtime.

The common-jet pair. For (9), solve the monotone scalar restricted equation by a bracketed root finder over $(-\epsilon, 0)$, for 41 logarithmically spaced $\epsilon \in [10^{-4}, 10^{-1}]$. Figure 2 plots the scaled inactive excess $(g_\sigma - 1)/\epsilon^2$. Its two curves remain on opposite sides of zero. Half the derivative separation divided by ϵ^2 ranges from 0.12362 to 0.12500, above the guaranteed constant $1/16$. This deterministic comparison displays the sign ambiguity while retaining the residual scale. Numerical root finding illustrates the construction; the exact inequalities and global convexity are proved in Appendix A.2.

7 LIMITATIONS

The certificate requires a justified full-Hessian variation bound on a specified active neighborhood. Obtaining a useful bound can be harder than computing a Hessian at the candidate. Small active eigenvalues enlarge the root radius and amplify coupling, so a mathematically valid test can abstain long after a solver has produced a useful approximation. The numerical study uses analytic bounds and exact constructed optima, making these effects inspectable while leaving large-scale implementation costs unresolved.

Strict margins are sufficient conditions. Boundary cases with $|q_j| + E_j = \lambda$ remain undecided, and a failed test does not imply a different optimal support. The formulation also assumes a nonempty candidate support and exact derivative information. Floating-point experiments are implementation checks; a machine-verifiable certificate would additionally require controlled rounding in eigenvalue bounds, linear solves, and interval comparisons.

The information obstruction is specific to one supplied jet and common bounds. It establishes a second-order rate with fixed constants on an explicit two-dimensional family, while the upper radius also depends on active curvature and cross-coordinate coupling. The results do not identify an optimal numerical constant or a lower bound for algorithms with additional evaluations. Finally, optimization support depends on the chosen objective and regularization strength; its agreement with a data-generating support requires separate statistical assumptions.

8 CONCLUSION

A restricted Newton correction can turn local curvature information into a signed-support certificate when explicit active and inactive margins exceed the remaining uncertainty. The cancellation in the inactive gradient explains the squared-residual radius, while a globally convex common-jet construction shows that order-two ambiguity can persist even under uniform strong convexity. The numerical study illustrates both successful certification and conservative abstention. A natural extension is to determine how a small number of additional derivative evaluations changes the residual scale required to resolve an inactive optimality threshold.

REFERENCES

- Francis Bach. Self-concordant analysis for logistic regression, 2009. URL <https://arxiv.org/abs/0910.4627>. arXiv.
- Antoine Bonnefoy, Valentin Emiya, Liva Ralaivola, and Remi Gribonval. Dynamic screening: Accelerating first-order algorithms for the lasso and group-lasso, 2014. URL <https://arxiv.org/abs/1412.4080>. arXiv.
- Cassio F. Dantas and Remi Gribonval. Stable safe screening and structured dictionaries for faster l1 regularization, 2018. URL <https://arxiv.org/abs/1812.06635>. arXiv.
- Cassio F. Dantas, Emmanuel Soubies, and Cedric Fevotte. Expanding boundaries of gap safe screening. *Journal of Machine Learning Research*, 2021. URL <https://www.jmlr.org/papers/v22/21-0179.html>.
- Olivier Fercoq, Alexandre Gramfort, and Joseph Salmon. Mind the duality gap: safer rules for the lasso. In *ICML*, 2015. URL <https://proceedings.mlr.press/v37/fercoq15.html>.
- Ankit Garg, Robin Kothari, Praneeth Netrapalli, and Suhail Sherif. Near-optimal lower bounds for convex optimization for all orders of smoothness, 2021. URL <https://arxiv.org/abs/2112.01118>. arXiv.
- Laurent El Ghaoui, Vivian Viallon, and Tarek Rabbani. Safe feature elimination for the lasso and sparse supervised learning problems, 2010. URL <https://arxiv.org/abs/1009.4219>. arXiv.
- Hiroki Ishizaka. Low-order cr-rt equilibrated-flux certification for semilinear problems on anisotropic meshes, 2026. URL <https://arxiv.org/abs/2602.01636>. arXiv.
- Tyler Johnson and Carlos Guestrin. Blitz: A principled meta-algorithm for scaling sparse optimization. In *ICML*, 2015. URL <https://proceedings.mlr.press/v37/johnson15.html>.
- Kwangmoo Koh, Seung-Jean Kim, and Stephen Boyd. An interior-point method for large-scale l1-regularized logistic regression. *Journal of Machine Learning Research*, 2007. URL <https://jmlr.org/papers/v8/koh07a.html>.
- Johan Larsson and Jonas Wallin. The hessian screening rule. In *Advances in Neural Information Processing Systems*, 2022. URL <https://arxiv.org/abs/2104.13026>.

-
- Jason D. Lee, Yuekai Sun, and Michael A. Saunders. Proximal newton-type methods for minimizing composite functions, 2014. URL <https://arxiv.org/abs/1206.1623>. SIAM Journal on Optimization.
- Sangkyun Lee and Stephen J. Wright. Manifold identification in dual averaging for regularized stochastic online learning. *Journal of Machine Learning Research*, 2012. URL <https://jmlr.org/papers/v13/lee12a.html>.
- Jingwei Liang, Jalal Fadili, and Gabriel Peyre. Activity identification and local linear convergence of forward-backward-type methods, 2015. URL <https://arxiv.org/abs/1503.03703>. arXiv.
- Mathurin Massias, Alexandre Gramfort, and Joseph Salmon. Celer: a fast solver for the lasso with dual extrapolation. In *ICML*, 2018. URL <https://proceedings.mlr.press/v80/massias18a.html>.
- Eugene Ndiaye, Olivier Fercoq, Alexandre Gramfort, and Joseph Salmon. Gap safe screening rules for sparsity enforcing penalties. *Journal of Machine Learning Research*, 2017. URL <https://jmlr.org/papers/v18/16-577.html>.
- Julie Nutini, Mark Schmidt, and Warren Hare. "active-set complexity" of proximal gradient: How long does it take to find the sparsity pattern?, 2017. URL <https://arxiv.org/abs/1712.03577>. arXiv.
- Neal Parikh and Stephen Boyd. Proximal algorithms. *Foundations and Trends in Optimization*, 2014. URL https://web.stanford.edu/~boyd/papers/prox_algs.html.
- Mark Schmidt, Nicolas Le Roux, and Francis Bach. Convergence rates of inexact proximal-gradient methods for convex optimization. In *NeurIPS*, 2011. URL <https://arxiv.org/abs/1109.2415>.
- Atsushi Shibagaki, Masayuki Karasuyama, Kohei Hatano, and Ichiro Takeuchi. Simultaneous safe screening of features and samples in doubly sparse modeling, 2016. URL <https://arxiv.org/abs/1602.02485>. arXiv.
- Adrien B. Taylor, Julien M. Hendrickx, and Francois Glineur. Smooth strongly convex interpolation and exact worst-case performance of first-order methods, 2015. URL <https://arxiv.org/abs/1502.05666>. arXiv.
- Jie Wang, Peter Wonka, and Jieping Ye. Lasso screening rules via dual polytope projection, 2012. URL <https://arxiv.org/abs/1211.3966>. arXiv.
- Zhen James Xiang, Yun Wang, and Peter J. Ramadge. Screening tests for lasso problems, 2014. URL <https://arxiv.org/abs/1405.4897>. arXiv.
- Lin Xiao. Dual averaging methods for regularized stochastic learning and online optimization. *Journal of Machine Learning Research*, 2010. URL <https://jmlr.org/papers/v11/xiao10a.html>.
- Hiroaki Yamada and Makoto Yamada. Dynamic sasvi: Strong safe screening for norm-regularized least squares. In *NeurIPS*, 2021. URL <https://proceedings.neurips.cc/paper/2021/hash/7b5b23f4aadf9513306bcd59afb6e4c9-Abstract.html>.

APPENDIX

A PROOFS OF THE MAIN RESULTS

A.1 SUPPORT CERTIFICATION

Proof. We prove Theorem 1. All norms in this proof are Euclidean norms or their induced operator norms. Let $P : \mathbb{R}^{|S|} \rightarrow \mathbb{R}^p$ be the coordinate embedding, so $P^T P = I$. Assumption 1 supplies convexity and differentiability, Assumption 2 supplies the local second-order control, and Assumption 3 will be used only after constructing an active stationary point. In particular, no existence of a global minimizer is assumed.

An auxiliary local root estimate. First suppose $a > 0$. On the closed ball $\mathcal{B} = \{h : \|h\| \leq 2a\}$, define

$$G(h) = P^T \nabla f(x + Ph) + \lambda s, \quad R_S(h) = G(h) - r - Hh.$$

The fundamental theorem of calculus, applied along the segment τh , gives

$$R_S(h) = \int_0^1 [P^T \nabla^2 f(x + \tau Ph) P - H] h \, d\tau, \quad \|R_S(h)\| \leq \frac{L}{2} \|h\|^2.$$

The segment lies in the assumed ball, so the Lipschitz bound applies throughout the integral. Since H is positive definite, $\|H^{-1}\| = 1/m$ and $\|d\| \leq a$. Define

$$\Phi(h) = -d - H^{-1} R_S(h).$$

For every $h \in \mathcal{B}$, the smallness condition $La/m \leq 1/4$ implies

$$\|\Phi(h)\| \leq a + \frac{L}{2m} (2a)^2 = a + \frac{2La^2}{m} \leq \frac{3}{2}a < 2a.$$

Moreover $DR_S(h) = P^T \nabla^2 f(x + Ph) P - H$, whose norm is at most $L\|h\|$. Integration along the segment joining any two points of the convex ball yields

$$\|\Phi(h) - \Phi(k)\| \leq \frac{2La}{m} \|h - k\| \leq \frac{1}{2} \|h - k\|.$$

For completeness, set $h_0 = 0$ and $h_{n+1} = \Phi(h_n)$. The self-map bound keeps every iterate in $\mathcal{B}$. Induction gives $\|h_{n+1} - h_n\| \leq 2^{-n} \|h_1 - h_0\|$. Hence for $N > n$,

$$\|h_N - h_n\| \leq \sum_{k=n}^{N-1} 2^{-k} \|h_1 - h_0\| \leq 2^{1-n} \|h_1 - h_0\|.$$

Completeness of finite-dimensional Euclidean space and closedness of the ball give a limit $h_* \in \mathcal{B}$. Continuity of Φ gives $h_* = \Phi(h_*)$, equivalently $G(h_*) = 0$. If k_* is another fixed point in this ball, the contraction inequality gives $\|h_* - k_*\| \leq \frac{1}{2} \|h_* - k_*\|$, hence equality of the two points. This proves the auxiliary root fact directly, including its existence assertion.

Newton error and local curvature. The root equation and the integral remainder give

$$h_* + d = -H^{-1} R_S(h_*), \quad \|h_* + d\| \leq \frac{L}{2m} \|h_*\|^2 \leq \frac{2La^2}{m} = b.$$

The self-map bound also improves the location to $\|h_*\| \leq 3a/2$. For all $h \in \mathcal{B}$, the Rayleigh quotient bound gives

$$P^T \nabla^2 f(x + Ph) P \succeq (m - L\|h\|)I \succeq (m - 2La)I \succeq \frac{m}{2}I.$$

Thus the curvature used below holds on a neighborhood of the constructed root, not just at the initial point. We have not needed global strong convexity.

A full-gradient remainder and cancellation. Define

$$R(h) = \nabla f(x + Ph) - \nabla f(x) - \nabla^2 f(x)Ph.$$

The full Hessian assumption, rather than an active-block assumption alone, is needed here. The same integral calculation gives

$$R(h) = \int_0^1 [\nabla^2 f(x + \tau Ph) - \nabla^2 f(x)]Ph d\tau, \quad \|R(h)\| \leq \frac{L}{2}\|h\|^2.$$

Since $R_S = P^T R$, the active root equation is $H(h_* + d) = -P^T R(h_*)$. For $j \notin S$, put $w_j = H_{jS}H^{-1}$. Substitution, before taking norms, yields the exact identity

$$\begin{aligned} \partial_j f(x + Ph_*) - q_j &= H_{jS}(h_* + d) + e_j^T R(h_*) \\ &= (e_j^T - w_j P^T)R(h_*). \end{aligned}$$

Because $P^T e_j = 0$, the two pieces of this row are orthogonal. Consequently

$$\|e_j^T - w_j P^T\|^2 = 1 + \|w_j\|^2, \quad |\partial_j f(x + Ph_*) - q_j| \leq 2La^2 \sqrt{1 + \|w_j\|^2} = E_j.$$

This proves the asserted output error with the stated constant. Bounding the two summands separately would lose this row-norm expression.

Signed support and optimality. Set $z = x + Ph_*$. For each $i \in S$, the coordinate error is no larger than its Euclidean norm, so

$$s_i z_i = s_i(x_i - d_i) + s_i(h_{*,i} + d_i) \geq s_i(x_i - d_i) - b > 0.$$

Thus z_S has sign s , and $z_{S^c} = 0$ by construction. For every inactive j , the output estimate and the strict margin give

$$|\partial_j f(z)| \leq |q_j| + E_j < \lambda.$$

Define $\eta_i = s_i$ for $i \in S$ and $\eta_j = -\partial_j f(z)/\lambda$ for $j \notin S$. The active equation gives $\nabla f(z) + \lambda\eta = 0$; the sign and margin facts give $\eta \in \partial\|z\|_1$. To see sufficiency explicitly, for every y convexity gives $f(y) - f(z) \geq \nabla f(z)^T(y - z)$, and coordinatewise convexity of absolute value gives $\|y\|_1 - \|z\|_1 \geq \eta^T(y - z)$. Adding the inequalities proves $F(y) \geq F(z)$. This constructs a global minimizer without assuming coercivity.

Global uniqueness from local information. Suppose y is another minimizer. In the preceding sum both the smooth supporting-plane slack and each absolute-value slack are nonnegative. Their sum is zero. For $j \notin S$, the coordinate slack is

$$|y_j| - \eta_j y_j \geq (1 - |\eta_j|)|y_j|.$$

Its strictly positive coefficient forces $y_j = 0$. Hence every minimizer lies on the same coordinate subspace. Convexity implies that $z + t(y - z)$, $0 \leq t \leq 1$, is an entire segment of minimizers. If $y \neq z$, choose $t_0 > 0$ small enough that for $0 \leq t \leq t_0$ the segment has active sign s and its active displacement from x stays below $2a$. Such a t_0 exists because all active coordinates of z are nonzero and $\|h_*\| \leq 3a/2 < 2a$. Along this segment the norm penalty is affine, and

$$\frac{d^2}{dt^2} F(z + t(y - z)) = (y_S - z_S)^T \nabla_{SS}^2 f(z + t(y - z))(y_S - z_S) \geq \frac{m}{2} \|y_S - z_S\|^2 > 0.$$

A constant function cannot have this positive second derivative. Thus $y = z$, proving global uniqueness.

Boundary cases and conclusion. If $a = 0$, then $r = d = b = 0$ and $E_j = 0$. Take $h_* = 0$ and $z = x$; the assumed margins directly give the preceding sign and KKT conditions. Uniqueness uses exactly the same slack argument. Continuity of $\nabla_{SS}^2 f$ and $H \succeq mI$ supplies a neighborhood of x with curvature at least $m/2$, replacing the zero-radius ball in the segment argument. If $L = 0$ and $a > 0$, both remainders vanish, Φ is constant, and $h_* = -d$; all estimates remain valid with zero output radius. If S is the whole coordinate set, inactive conditions and slack statements are empty; the curvature segment argument alone establishes uniqueness. Equality in an inactive or active margin was not assumed and is not covered by the strict-support conclusion. Therefore the point constructed above is the unique minimizer z^* and satisfies every bound and signed-support assertion of Theorem 1. $\square$

A.2 THE COMMON-JET CONSTRUCTION

Proof. We prove Theorem 2, including uniform regularity constants and the precise information restriction. Fix $0 < \epsilon \leq 1/10$, $\kappa = 1/8$, $B = 9/64 = 9\kappa^2$, and $\sigma \in \{-1, +1\}$. Write $w = (t, v)$, $t = u - 1$, and $\rho = \|w\|$, with f_σ as in (9). Each function is a polynomial, so every derivative used below exists globally.

An auxiliary convexification estimate. The Hessian of the cubic term $\sigma\kappa t^2 v$ is

$$C_\sigma(w) = \sigma\kappa \begin{pmatrix} 2v & 2t \\ 2t & 0 \end{pmatrix}.$$

Its operator norm is bounded by its Frobenius norm, which satisfies

$$\|C_\sigma(w)\| \leq \kappa \sqrt{4v^2 + 8t^2} \leq 6\kappa\rho.$$

The quartic term has Hessian $4B\rho^2 I + 8Bww^T$, whose second summand is positive semidefinite. Hence for every w ,

$$\nabla^2 f_\sigma(u, v) \succeq (1 - 6\kappa\rho + 4B\rho^2)I = \left(1 - \frac{9\kappa^2}{4B} + 4B\left(\rho - \frac{3\kappa}{4B}\right)^2\right)I \succeq \frac{3}{4}I.$$

The deliberately loose cubic bound is sufficient because $B = 9\kappa^2$. This proves global strong convexity with the same lower bound for both signs. Integrating the Hessian lower bound along rays gives a lower bound of degree two on f_σ , so adding $|u| + |v|$ produces a coercive, continuous, strictly convex objective. A minimizing sequence is bounded by coercivity; a convergent subsequence and continuity give existence. Strict convexity gives a unique minimizer.

Common information and a uniform Hessian modulus. At $x = (1, 0)$, we have $w = 0$, so the cubic and quartic terms and their derivatives through order two vanish. Direct differentiation gives

$$f_\sigma(x) = 0, \quad \nabla f_\sigma(x) = \begin{pmatrix} \epsilon - 1 \\ 1 - \epsilon^2/16 \end{pmatrix}, \quad \nabla^2 f_\sigma(x) = I.$$

In particular the active set $S = \{1\}$, active sign $+1$, curvature $m = 1$, residual $r = \epsilon$, and the complete value-gradient-Hessian information are identical.

For the asserted common Hessian-Lipschitz constant, let h be a direction. The derivative of the cubic Hessian has norm at most $6\kappa\|h\|$. Differentiating the quartic Hessian gives

$$D(4B\rho^2 I + 8Bww^T)[h] = 8B(w^T h)I + 8B(hw^T + wh^T),$$

whose norm is at most $24B\rho\|h\|$, by Cauchy-Schwarz and the rank-one norm identity $\|ab^T\| = \|a\|\|b\|$. Thus on $\rho \leq 1/2$,

$$\|D\nabla^2 f_\sigma(w)[h]\| \leq (6\kappa + 12B)\|h\| = \frac{39}{16}\|h\| < 4\|h\|.$$

The ball is convex, so integrating this derivative along any chord proves the claimed full Hessian Lipschitz bound $L = 4$. Both signs therefore supply the same valid constants even if those constants, the radius $1/2$, and the global curvature lower bound are included in the available information.

The common restricted minimizer. On $v = 0$, the two objectives coincide. Near $u = 1$, the derivative of $f_\sigma(u, 0) + |u|$ is $\psi(t) = t + 4Bt^3 + \epsilon$. We have $\psi'(t) = 1 + 12Bt^2 > 0$, $\psi(-\epsilon) = -4B\epsilon^3 < 0$, and $\psi(0) = \epsilon > 0$. The intermediate value theorem and strict monotonicity give exactly one root $t_\epsilon \in (-\epsilon, 0)$. It has $u_\epsilon = 1 + t_\epsilon > 9/10$, so the differentiation of $|u|$ was valid. Because the entire restricted objective is strictly convex, this stationary point is its unique global minimizer, including over $u \leq 0$.

Put $a_\epsilon = |t_\epsilon|$, which is a scalar local to this proof. The root equation gives

$$\epsilon = a_\epsilon(1 + 4Ba_\epsilon^2), \quad \frac{\epsilon}{1 + 4B\epsilon^2} \leq a_\epsilon < \epsilon.$$

The upper bound uses $a_\epsilon < \epsilon$; substituting it into the denominator gives the lower bound. Since $1 + 4B\epsilon^2 \leq 1609/1600 < \sqrt{2}$, we obtain the uniform quantitative estimate

$$\frac{\epsilon^2}{2} < t_\epsilon^2 < \epsilon^2.$$

These inequalities justify all subsequent threshold and separation comparisons without an asymptotic remainder.

Different support decisions. At the common restricted minimizer $z_\epsilon = (1 + t_\epsilon, 0)$, the inactive smooth gradient is

$$g_\sigma = \partial_v f_\sigma(z_\epsilon) = 1 - \frac{\epsilon^2}{16} + \sigma \kappa t_\epsilon^2.$$

For $\sigma = +1$, the lower bound just proved yields $g_+ > 1 - \epsilon^2/16 + (1/8)(\epsilon^2/2) = 1$. The inactive KKT condition for penalty parameter one is therefore violated. If the full minimizer had $v = 0$, its first coordinate would necessarily minimize the restricted objective and hence equal $1 + t_\epsilon$, which has just been ruled out. Thus the full minimizer for the plus case has nonzero second coordinate.

For $\sigma = -1$, we have $g_- < 1$ and

$$g_- > 1 - \frac{\epsilon^2}{16} - \frac{\epsilon^2}{8} = 1 - \frac{3\epsilon^2}{16} > -1.$$

The active derivative of the regularized objective vanishes at $u_\epsilon > 0$, and the inactive gradient lies strictly inside $[-1, 1]$. The coordinate subgradient inequalities and convexity, as in the first proof, now prove full optimality of z_ϵ . Global strict convexity proves uniqueness. Its support is exactly $\{1\}$, whereas the plus case includes coordinate two. This is the required support ambiguity.

The scalar information lower bound. A deterministic estimator whose input consists only of the common value, gradient, Hessian, candidate point, penalty parameter, and common regularity constants receives identical data for the two functions. It must therefore return the same scalar q . The triangle inequality gives

$$2 \max\{|q - g_+|, |q - g_-|\} \geq |g_+ - g_-| = 2\kappa t_\epsilon^2 > \frac{\epsilon^2}{8}.$$

Consequently its worst-case absolute error is at least $\epsilon^2/16$. A deterministic binary support decision must likewise give the same decision for both inputs, and so fails on at least one of them. A rule permitted to abstain is consistent with this statement. A procedure permitted another derivative query has different information and is not covered by this argument.

Boundary checks and conclusion. The estimates hold uniformly throughout $0 < \epsilon \leq 1/10$; no limit exchange is needed. At $\epsilon = 0$ the separation vanishes, which is why that endpoint is excluded. The local regularity ball contains the restricted root because $|t_\epsilon| < 1/10 < 1/2$. For $\epsilon \leq 1/16$, even the smallness condition $L\epsilon/m \leq 1/4$ of Theorem 1 holds; the two functions need not satisfy its inactive margin condition. Thus the construction does not contradict the certificate. It establishes an order- ϵ^2 obstruction for this precise common-jet information class, not an oracle lower bound for iterative algorithms. Therefore all regularity, support, and estimation assertions in Theorem 2 follow. $\square$